



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

COMPILER DESIGN-CSA1402 CO1_AT2 ERROR ANALYSIS

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Problem 1:

The following program is intended to perform simple arithmetic operations, function calls, array access, and looping constructs. However, the code contains multiple errors introduced at different phases of compilation, including lexical, syntax, semantic, logical, and runtime errors.

The screenshot shows a C++ IDE with a file named `sample1.cpp`. The code is as follows:

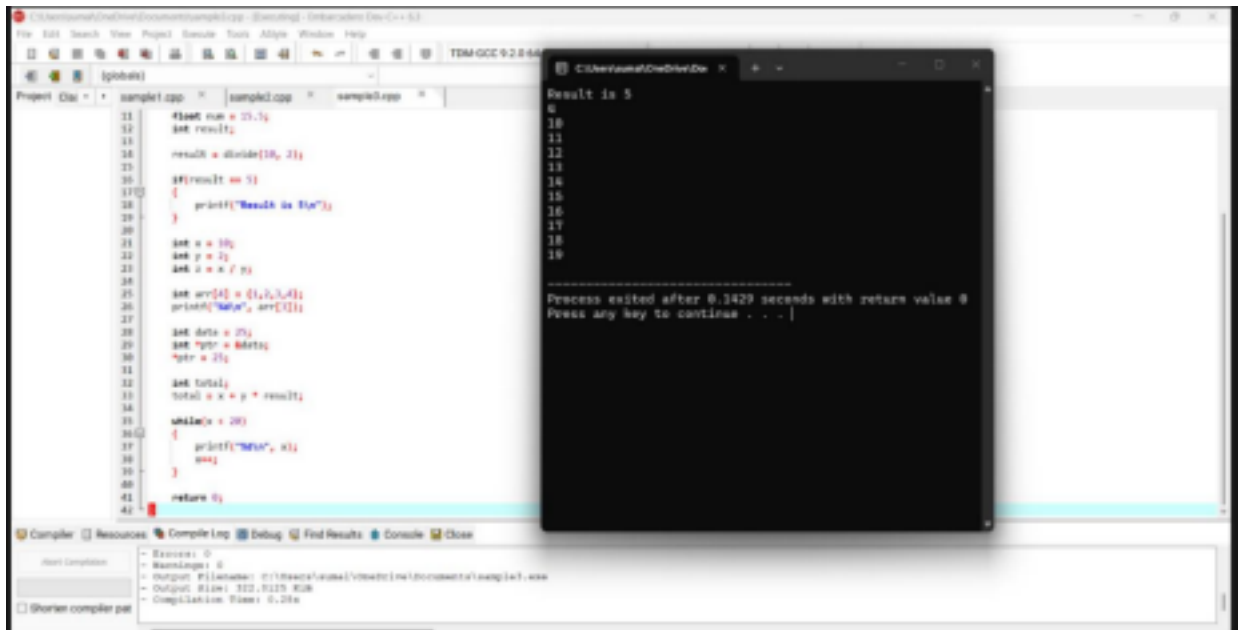
```
1 // Header files
2 #include <iostream>
3 using namespace std;
4
5 int sum = 0;
6
7 fun(i = 0, j = 10) {
8     sum = sum + arr[i];
9 }
10
11 return sum;
12
13
14 int main() {
15     int arr[10] = {1,2,3,4,5,6,7,8,9,10};
16     int total;
17     total = sumfun(arr, 10);
18     cout << total << endl;
19     return 0;
20 }
```

The output window shows the following text:

```
Correct Sum
Division = 5
Pointer Value = 20
Process exited after 0.126 seconds with return value 0
Press any key to continue . . .
```

Problem 2:

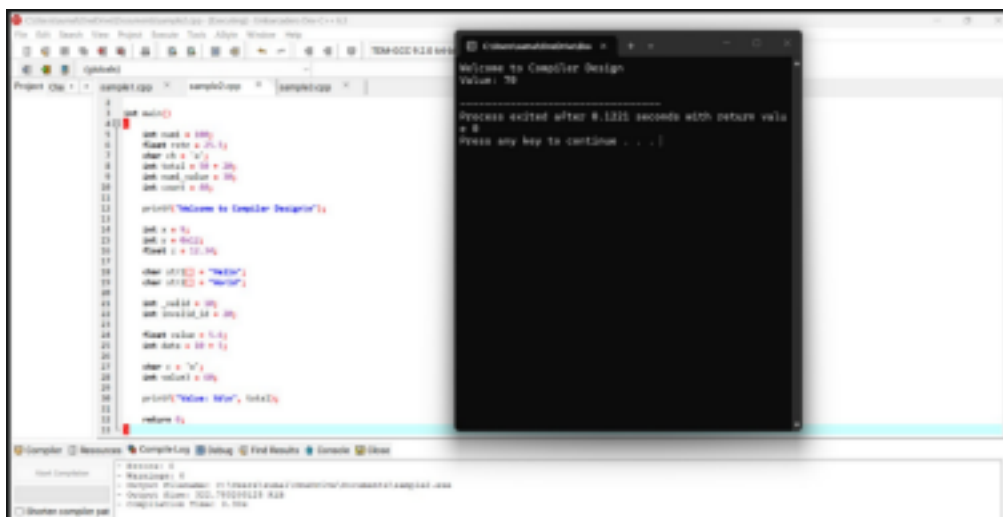
The following C program contains multiple lexical errors introduced intentionally. These errors occur due to invalid tokens, incorrect identifiers, malformed constants, and illegal symbols, which are typically detected during the lexical analysis phase of a compiler. Carefully examine the program and answer the questions given below.



Problem 3:

The following C program is designed to perform arithmetic operations, function calls, conditional checks, pointer usage, and looping. However, the program contains multiple errors distributed across different phases of a compiler, including:

- Lexical Errors
- Syntax Errors
- Semantic Errors
- Intermediate Code Generation Issues (logic/precedence)
- Code Optimization Issues (redundancy/inefficiency)
- Runtime / Target Code Errors



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📁 lex programs	Create README.md		now
📄 Exp 1_192521302.png	Add files via upload		last week
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📄 Lex 1.png	Add files via upload		now
📄 Lex2.png	Add files via upload		now
📄 Lex3.png	Add files via upload		now
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📄 Lex5.png	Add files via upload		now
📄 Lex6.png	Add files via upload		now
📄 Lex7.png	Add files via upload		now
📄 error 1.png	Add files via upload		now
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📄 error3.png	Add files via upload		now